Motion & graphs

Describe journeys

1 Distance vs displacement

Distance: total path, scalar.
Displacement: straight start-to-finish change with direction, vector.

Round trip → zero displacement, nonzero distance. Average speed = total distance / total time.

2 Useful estimates

Walking ≈ 1.5 m/s; running 3; cycling 6; sound in air 330. Conditions change values.

$\mathbf{s = vt}$ at constant speed. Convert minutes to seconds and km to m before calculating.

Calculate changes

3 Velocity & acceleration

Velocity = speed with direction. Circle: constant speed can still mean changing velocity.

$\mathbf{a = (v-u)/t,}$ m/s^2. 5 → 17 m/s in 4 s gives 3 m/s^2.

4 Uniform acceleration

$\mathbf{v^2-u^2 = 2as}$ only for constant a. From rest, a=2 over 25 m gives v=10 m/s.

Free fall near Earth ≈ 9.8 m/s^2 when air resistance is negligible.

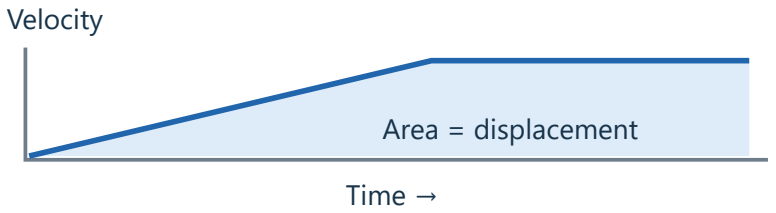
Read slopes & areas

5 Distance–time

Gradient = speed. Flat = rest; straight slope = constant speed.

Curved graph: draw tangent at required time; gradient estimates instantaneous speed. Use a large gradient triangle.

6 Velocity–time



Gradient = acceleration. Flat = constant velocity. Signed area = displacement; sum absolute areas for distance.

Triangle + rectangle areas give movement. Count squares for a curved section if needed.